

Global Player Mobility in Football: A League-to-League Network Analysis

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Domain-specific lecture: *Foundations of International Business* (WU Wien)*

June 15, 2026

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Abstract

This project analyzes global player mobility in professional football using transfer data from the Transfermarkt website. Player moves are represented as a directed, weighted network on league level, where edges capture transfer flows between source and destination competitions across seasons. Four structural hypotheses were tested, concerning market concentration (H1), the bridging role of mid-tier leagues (H2), community structure relative to a degree-preserving null model (H3), and the domestic orientation of lower-tier transfers (H4). The market is found to be only moderately concentrated and becoming less so over time, mid-tier leagues do not act as disproportionate bridges, community structure is strong and geographically coherent, and lower-tier transfers are overwhelmingly domestic. The work follows CRISP-DM and is fully reproducible.

Motivation

Professional football is a global labor market, where players move between clubs and leagues across countries. According to FIFA's global Transfer Report 2025, clubs worldwide completed 24,558 international transfers in men's professional football during 2025, USD 13.08 billion, an increase of more than 50% on the previous year. These figures show that this labor market is large, volatile and structurally embedded in the game of football.

Despite this scale, public discourse often uses qualitative measures about individual highly paid moves and some recurring labels such as “selling leagues”, “import leagues”, or “stepping-stone pathways”. These labels are rarely quantitative or defined properly. From a data-science perspective the transfer market is a natural candidate for a network analysis. Each transfer can

*Completion date / certificate issue date to be inserted once the certificate is issued.

be modeled as a directed event from a source club to a destination club, embedded in a source and destination league, with an exact timestamp. This structure forms a directed, weighted network which can be used to extract valuable information about structural roles of leagues, the geography of player flows, and the temporal evolution of market concentration.

Problem Statement & Research Questions

The goal of this project is to model the global transfer market as directed, weighted network at the league level, and to use this network and appropriate tools to answer four research questions about its structural properties. The unit of analysis is the league (competition), each node is a single league, each directed edge(u, v) represents player transfers from league u to league v , and edge weights are transfer counts aggregated over a defined time window. The datasource is Transfermarkt (www.transfermarkt.com) and the analysis covers 32-seasons, from 1994/95 to 2025/26. The four research questions and hypothesis are as follows:

- **RQ1 – Mobility structure and concentration.** How is player mobility between leagues structured as a directed and weighted network, and how concentrated are incoming transfer flows across leagues?
H1: Incoming transfer flows are highly concentrated, with a small set of leagues accounting for a large share of all incoming transfers.
- **RQ2 – Centrality and roles.** Which leagues primarily act as senders or receivers of players, and which leagues function as bridges between otherwise weakly connected parts of the network?
H2: Some mid-level leagues exhibit disproportionately high betweenness centrality and act as bridges between regions and/or tier systems.
- **RQ3 – Communities and mobility regions.** Are there clusters of leagues that form strongly connected mobility ecosystems, and how stable are these clusters across seasons?
H3: The network exhibits non-random community structure, with detected communities showing higher modularity than a degree-preserving randomised baseline.
- **RQ4 – Lower-league pathways.** How important are transfers between lower tiers and top tiers, and how do these pathways differ from top-tier mobility?
H4: Transfers originating from lower tiers are more likely to be domestic and upward (from lower to higher tier) within the same country, while cross-border moves are relatively less common compared to top-tier mobility.

The Herfindahl–Hirschman Index and Top-5 inflow share for H1, the mid-tier share among the top-25 leagues by betweenness centrality (and the lift relative to the mid-tier base rate) for H2, the empirical p -value of observed modularity against a degree-preserving null model for H3, and the weighted domestic share of outgoing transfers by source-tier group for H4. The full KPI definitions and thresholds are reproduced in Section 3.

The project is primarily a descriptive and structural analysis and does not attempt to predict transfer feed, forecast individual moves or model player valuations. The contribution here is to characterise the structure of the global league-to-league network, to test the four hypothesis against it and do so reproducibly across multiple season over almost 30 years.

0.1 Outline

This report closely follows the CRISP-DM process model [1]. Section 1 reviews prior work on football transfer networks and situates this project’s methodological choices within that literature. Sections 3 to 8 correspond to the six CRISP-DM phases: business understanding (Section 3), data understanding (Section 4), data preparation (Section 5), modelling (Section 6), evaluation (Section 7), and deployment (Section 8). Section 9 concludes with a verdict-by-hypothesis summary and a brief discussion of limitations.

All stochastic steps in the analysis are seeded with the author’s student ID (seed = 12412672) for full reproducibility, and the analysis pipeline is split across two notebooks executed in sequence under a pinned Python environment.

1 Related Work

Work on football Transfermarkt data can be categorized in two different ways. The first focuses on individual player valuation typically using regression or machine-learning models to predict fees or future prices while the second strand focuses on network analysis of transfer flows. This project focuses on the second strand is positioned as follows.

Earlier work on networks operate mostly on club-level. [2] cover 24 top-class leagues worldwide between 2011 and 2015 and relate club centrality to sportive success. [3] build a club-level network of roughly 470,000 transfers over 1990–2016, involving more than 23,000 clubs across 206 countries and characterize its degree and strength distributions. [4] cover 21 European top-tier leagues over 1996/97–2015/16 and combine network metrics with self-organising maps. Work like [5] add depth by analyzing fee-weighted networks of eight highest-tier european leagues from 1992 – 2020. They find that the market is not small-world, maybe because clubs primarily trade with financially similiar partners and they explicitly mention that follow-up work should replace fees with transfer counts as edge weights to also include lower-tier leagues. [6] run community detection on the seven top European leagues between 2009/10–2021/22 under both weightings and report the fee-weighted cluster the richest clubs cross-nationally, where the count-weighted edges "faithfully reconstruct the national leagues".

This project differs from the above in several ways. First, the unit of analysis is the league rather than the club, which shows patterns on a national level. Second, the scope covers all professional leagues recorded on Transfermarkt over 32 seasons (1994/95–2025/26), including all lower-tier and academy leagues. Third, the edges are weighted by transfer counts rather than fees.

Methodologically the community detection uses Leiden algorithm, which improves the Louvain algorithm by guaranteeing well-connected communities. [7]. Significance is tested against a degree-preserving null model implemented via random edge rewirign. The remporal trends are tested with the non-parametric Mann–Kendall test. [8] ? ?

The combination of a league-level, global, count-weighted, multi-decade scope with Leiden-based community detection and a null-model significance test is, to the knowledge when writing this report, novel.

2 Methodology

2.1 Process model: CRISP-DM

The project closely follows the CRISP-DM process (?). The six phases map to the remaining sections of this report as follows: business understanding (Section 3), data understanding (Section 4), data preparation (Section 5), modelling (Section 6), evaluation (Section 7), and deployment (Section 8). The proposal’s research questions and hypotheses (H1–H4) and the corresponding KPIs (Table 1) were fixed at the business-understanding phase and carried through unchanged.

2.2 Methods stack

Network structure is described with standard measures on a directed, weighted graph with measures like weighted in/out strength and distance-weighted betweenness centrality. Community structure is detected with the Leiden algorithm ?. Statistical inference uses the Herfindahl–Hirschman Index (HHI) for concentration which measures market concentration with 0 being perfect equality and one being a full monopoly, a degree-preserving null model for the significance of observed modularity, and the non-parametric Mann–Kendall trend test (??) for temporal trends. The domain-specific lecture *Foundations of International Business* (WU Wien) frames the transfer market as a structured cross-border labour market and motivates the geographic and tier-based hypotheses H3 and H4.

2.3 Experiment design and reproducibility

All stochastic steps are seeded with the author’s student ID, $\text{seed} = 12412672$. The pipeline is split into two notebooks run in sequence: `01_data_preparation.ipynb` produces the cleaned edge lists under `data/prepared/`, and `02_analysis.ipynb` uses them and writes all figures and tables to separate folders. The dataset itself is private.

Robustness is addressed by computing the H1 concentration metrics in four variants (with and without self-loops, with and without loan transfers) and by computing H1 and H3 metrics per season to check temporal stability.

All experiments and code can be found at:

<https://github.com/timg4/Mobility-in-Global-Football>

3 Business Understanding

The goal of this project just as described in the proposal is to characterise the structure of the global league-to-league transfer network and test four hypotheses about its concentration, bridging behaviour, community structure, and the domestic orientation of lower-tier transfers. Success is defined ex ante by the KPIs and decision thresholds in Table 1; the results are evaluated against these in Section 7.

Methodological choices (league-level unit of analysis, count-weighted edges, exclusion of pseudo-destinations) and the reproducibility setup (seeded randomness, pinned environment, two-notebook pipeline) are described where they apply, in Sections 5–8.

H	KPI	Decision rule
H1	HHI of inflows; Top-5 league share; Mann-Kendall trend test per season	“Concentrated” if $HHI > 0.15$ or Top-5 share $> 50\%$; trend significant at $\alpha = 0.05$.
H2	Mid-tier share (tier 2–4) among top-25 leagues by betweenness, vs. mid-tier base rate	“Supported” if lift $\geq 1.5\times$.
H3	Observed modularity Q vs. degree-preserving null model	$p < 0.05$ one-sided.
H4	Weighted domestic share of outgoing transfers by source-tier group	“Supported” if domestic share $> 70\%$ for tier-3 and lower.

Table 1: KPIs and ex-ante decision rules for hypotheses H1–H4.

4 Data Understanding

4.1 Source

The data is directly sourced from Transfermarkt.com and was provided by WU Wien. It contains five tables in total from which three are relevant for this project, the transfers, player and club table. Among other information, each transfer record links a player, a source and destination club, source and destination competition (league) identifiers, source and destination country identifiers, a transfer type, a fee field, and a date.

4.2 Missing values& initial observations

- **Missing competition IDs.** A large share of transfer records (300.000 transfer records) have a missing source and/or destination competition ID while not a single club ID is missing in the entire dataset.
- **Pseudo-destinations-encoding** Looking for the clubs with the most in-/out transfers revealed that several club-ids are in fact non-club states with a specific encoding used by Transfermarkt. club 123 denotes *Retired*, club 515 *Without Club*, club 75 *Unknown Club*, and club 2113 *Career Break*.
- **Missing fees.** The fee field is missing for a large fraction of records (880.000 transfer records). The fee-label field revealed that most of these are genuine free transfers but transfer fees in global football are also frequently undisclosed. Missing fees are therefore not missing at random.
- **Encoded dates.** The date field is an integer. Three candidate encodings (days, seconds, milliseconds since 1970-01-01) were tested against a plausible date window. This was done by comparing with three well-known transfers: Neymar to Paris Saint-Germain (2017), Cristiano Ronaldo to Real Madrid (2009), and Ronaldo to Inter (1997). All of them were mapped correctly when using days since 1970-01-01.

4.3 Temporal coverage

The transfers are sparse and incomplete before the mid mid-1990s in our dataset. It has some implausible outliers in the earliest years and some record dated after the dataset handover. (from which most are planned loan returns). The coverage for each year revealed that the data becomes reliable from 1995 onwards, so the analysis window was fixed from 1 January 1995 to 27 January 2026. Everything before that was dropped due to sparsity and everything after because planned returns from loans are unsure and therefore should not be mapped as transfer yet.

4.4 Descriptive overview

After cleaning and restriction to the analysis window, the dataset has the figures that can be seen in Table 2.

A total of 921,332 cleaned transfer records fall within the analysis window of which 586,520 have an identified league at both ends and are therefore usable for this analysis. The strict subset spans 672 distinct leagues across 118 countries. Loans account for roughly one in four transfers.

Dimension	Value
Analysis window	1994/95 – 2025/26
Seasons	32
Transfer records (clean, in window)	921,332
Records with both leagues mapped (strict)	586,520
Unique leagues (strict)	672
Unique countries (strict)	118
Loan transfers	249,985 (27.1%)
Permanent transfers	671,347 (72.9%)

Table 2: Descriptive overview of the cleaned dataset.

4.5 Transfer types and fees

The transfer-type field shows among others permanent moves, loans, loan returns and free transfers. Loans are legitimate form of transfers and are therefore included in the analysis. The non-random pattern of missing fees seems to be systematic and using them in the analysis would be biased against smaller and non-European leagues. Therefore in the next phases the network edges are weighted by transfer counts and not fees and fees are not included further in the analysis.

5 Data Preparation

The data preparation produces four edge-list files by following the steps below in order.

5.1 Date decoding and window restriction

The integer date field is decoded as days since 1970-01-01 (Section 4). Records before 1 January 1995 are dropped as pre-coverage, and records after the dataset handover (27 January 2026) are dropped as out-of-window. Each remaining transfer is assigned a season using a July cut-off: transfers from July of year Y through June of year $Y+1$ belong to season $Y/Y+1$.

5.2 Pseudo-destination handling

The four club IDs identified that do not correspond to real clubs were rewritten to readable pseudo-code (RETIREMENT, WITHOUT_CLUB, UNKNOWN_CLUB, CAREER_BREAK) so these record can be tracked through the pipeline rather than them just being discarded.

5.3 Conservative competition imputation

Many transfers had a missing competition ID although both club IDs were known. For each club we look at all the competition IDs in other transfers. If there is only one competition ID, we use that one. If it has ambiguous competition IDs, we leave it out. Records whose competition cannot be recovered are labeled UNKNOWN_COMPETITION.

5.4 Strict and with-unknown variants

Two parallel variants of the cleaned data are produced:

- `with_unknown`: retains transfers whose source or destination competition is UNKNOWN_COMPETITION or a pseudo-destination, labelled explicitly.
- `strict`: drops all transfers where either endpoint is unidentified (UNKNOWN_COMPETITION or pseudo-code).

All analyses in this report use the `strict` variant.

The unknown and pseudo endpoints cannot be used meaningfully as nodes in the league-level network and keeping them in the network would inflate the degree and strength of artificial nodes. The volume of identified league-to-league transfers (586,520, see Table 2) is sufficient to support all downstream analyses.

5.5 Edge-list aggregation

For each variant, transfers are aggregated to a directed league-to-league edge list. Two granularities are produced:

- `edge_all_*`: aggregated over all 32 seasons; one row per (source league, destination league) pair.
- `edge_season_*`: per-season; one row per (source league, destination league, season) triple.

Each edge row carries the source and destination league IDs and names, country IDs, tier classifications, the number of transfers (`n_transfers`), the number of loans (`n_loans`), fee sum and median, and the domestic share (the fraction of transfers on that edge with identical source and destination country).

5.6 Competition metadata

A supplementary mapping file (`competition_mapping.csv`) maps each competition ID to a human-readable league name, country, tier (1–4, plus a Youth category), canonical ID for renamed leagues, and prior league names. This mapping from leagues that were renamed over time was done by hand using the Transfermarkt.com website directly. This file is produced by a separate scraping script and the mapping of leagues that changed their names over time was done by hand using the Transfermarkt.com website directly. This is required for tier-based analyses (H2, H4) and for the readable league labels in all figures.

6 Modelling

All modelling is implemented on the strict edge-list variant.

6.1 Graph construction

The aggregated directed weighted graph G_{all} is built from the strict file that contains all edges and aggregates over all seasons. It uses `igraph`, with one node per unique league ID and one edge per (source $\rightarrow$ destination) league pair. The edge weight is the number of player transfers over the 32-season window. A second edge attribute `distance = 1/weight` was also added so that the betweenness-centrality computations interpret heavy flows as short paths. For node labels real league-names were used. The graph G_s was constructed per season in an identical way to analyze the temporal hypotheses H1 and H3. Self-loops, so tranfers that happened within one league, were kept in the graph but excluded when computing the cross-league metrics. For the null-model rewiring in H3 these loops were remove because they cannot be preserved under degree-preserving edge rewiring.

6.2 Node-level network metrics

For each league in G_{all} the following metrics are computed: weighted in- and out-strength (sum of incoming/outgoing edge weights), in- and out-degree (number of distinct partner leagues), and directed weighted betweenness centrality on the `distance` attribute. To rank the betweenness, shortest paths on inverse weights were used, so that heavy corridors contribute short edges.

6.3 Community detection

To detect communities within the graph the Leiden algorithm (?) was used with the `RBConfigurationVertexPartition` quality function and weighted by the number of transfers. The RB-configuration partition can be used for weighted directed graphs and allows resolution tuning. It is also used ofr descriptive community structure such as community labels or a meta-graph.

6.4 Null-model

To test the significance of H3 adjustments were necessary make the observed and null modularity comparable.

The quality function is switched to `ModularityVertexPartition` which gives the standard modularity Q on a scale that is directly comparable across graphs of similar size. Also self-loops are removed since the degree-preserving rewiring is not defined on loops. The observed modularity Q^{obs} is then computed on the self-loop-free graph using Leiden with that quality function.

The null distribution is generated by 20-degree-preserving random rewirings of the graph. Each rewiring shuffles the edge endpoints of each node (with in- and out-degree staying the same) and a rewiring factor of two times the edge count. Due to computational limitations the permutations were reduced from the originally planned 1000 to 20. The gap between the observed and the null modularity is already really large so that no realistic null sample would change that conclusion. Leiden is then used on all rewired graphs and resulting in Q_i^{null} for each run. The one-sided p -value then is $p = \#\{i : Q_i^{\text{null}} \geq Q^{\text{obs}}\} / 20$.

The modularity per-season is computed in the same way on each G_s to check the temporal stability of these communities. This temporal analysis is limited to the seasons from 2000/01 onwards, where the coverage is dense enough.

6.5 Hypothesis-specific computations

H1 – Concentration. The HHI of inflows is computed as $\text{HHI} = \sum_v (s_v^{\text{in}} / S)^2$ over all destination leagues v , where s_v^{in} is the weighted in-strength and S the total inflow.

The Top-5 share is the sum of the five largest s_v^{in} divided by S . Both are computed on the aggregated graph and per season. The Mann–Kendall test is applied to the two per-season series to test for a monotonic trend over time.

H2 – Bridge leagues. Leagues are ranked by directed weighted betweenness centrality on distance. Mid-tier is defined as tier 2–4. Among the top-25 leagues by betweenness, the mid-tier share is compared to the mid-tier base rate across all leagues in the graph. This ratio is the *lift*.

H4 – Lower-league pathways. Source leagues are grouped into four tiers: Top (tier 1), Second (tier 2), Mid (tier 3–4), Lower (tier 5+). For each group the weighted domestic share is computed as a transfer-count-weighted mean of the per-edge `domestic_share`.

7 Evaluation

In this section the actual results are reported and each hypothesis is evaluated against the KPI thresholds defined Table 1. This section reports the results obtained on the aggregated and per-season graphs and evaluates each hypothesis against the KPI thresholds defined in Table 1. All numbers are computed on the strict edge-list variant.

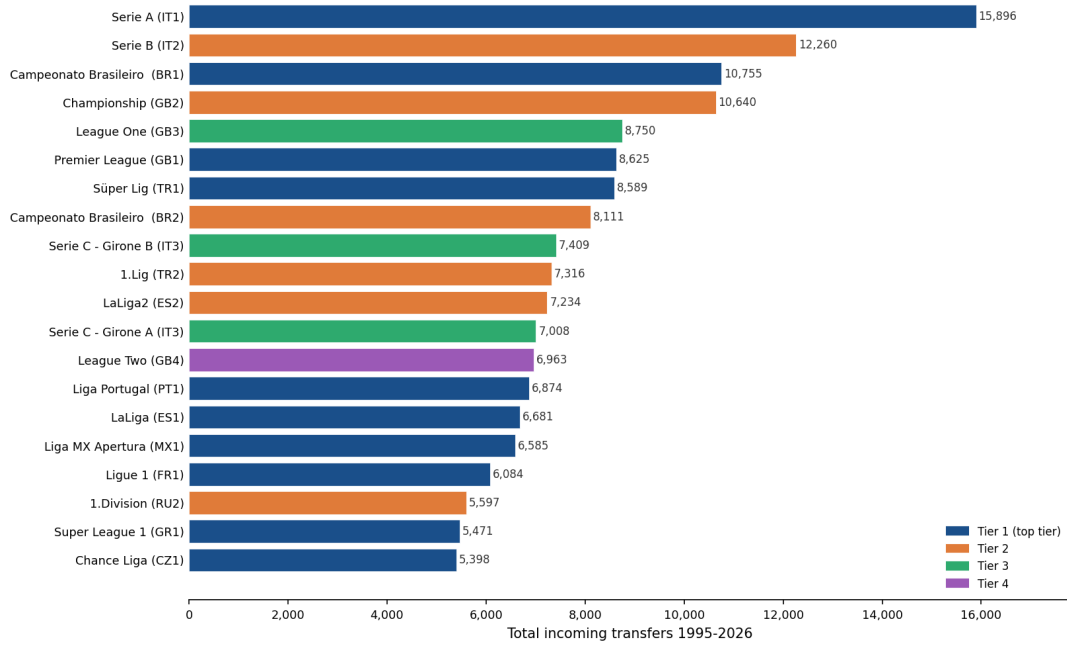


Figure 1: Top 20 leagues by incoming transfers

7.1 H1 – Concentration

On the aggregated graph the HHI of inflows is $HHI = 0.0094$, far below the 0.15 threshold for “concentrated”. The top-5 leagues receive 12.3% of all cross-league inflows, and the top-10 receive 21.0%, both well below the 50% threshold. The largest single inflow destination are Serie A (Italy, 16,002 transfers), Serie B (Italy, 12,539), the Championship (England, 11,514), Premier Liga (Ukraine, 11,046) and Campeonato Brasileiro Série A (Brazil, 10,988); see Figure 1

The Mann-Kendall test rejects the null of no monotonic trend strongly for both metrics, with the HHI showing a decreasing trend with $\tau = -0.93$, $p < 10^{-4}$ and the Top-5 share decreasing with $\tau = -0.93$, $p < 10^{-4}$. This means that the Top-5 share of incoming transfers declines from roughly 27% in 2000/01 to roughly 9% in 2024/25.

Verdict. *H1 is not supported.* The market therefore is unequal but not concentrated by the thresholds but there is a very interesting temporal finding. The concentration falls monotonically across the entire observation window.

7.2 H2 – Bridge leagues

Initial result. Directed weighted betweenness centrality on the aggregated graph produces a ranking dominated by large top-tier leagues. Among the top-25 leagues by betweenness, the mid-tier (tier 2–4) share is 40.0%, compared to a mid-tier base rate of 40.3% across all 672 leagues. (see Figure 2) The resulting lift is $0.99\times$, essentially identical to chance and well below the $1.5\times$ threshold for “supported”.

CRISP-DM iteration Further inspection of this result during evaluation phase showed a methodological weakness. The betweenness centrality like it was used in this project measures the topological shortest-path intermediation rather than the actual bridge role that was searched for. A bridge would rather receive players from one region and send them to another. Inspection of this result during the evaluation phase exposed a methodological limitation: betweenness centrality captures topological shortest-path intermediation rather than the brokerage role envisioned in the proposal – namely, a league that receives players from one region and sends them on to another. In line with the iterative nature of CRISP-DM, the project returned to the modelling phase (Section 6) and computed three additional operationalisations of the bridge concept before re-evaluating the hypothesis.

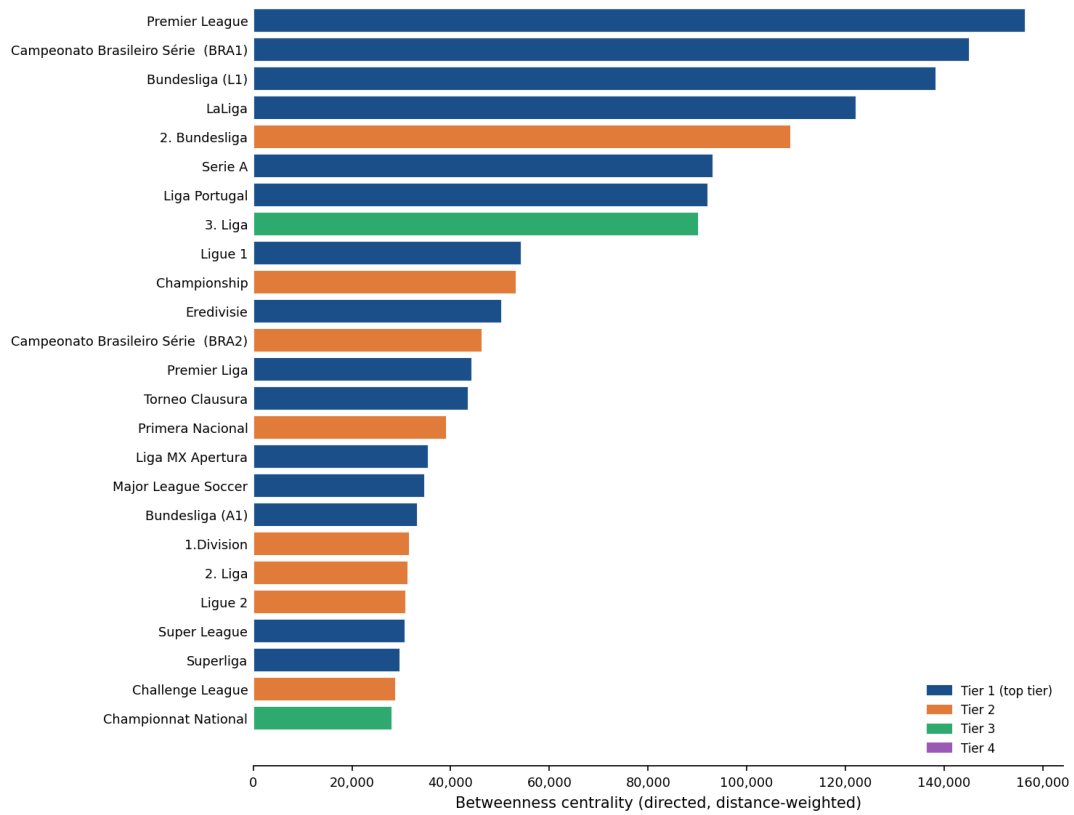


Figure 2: Top 25 leagues by betweenness centrality

Verdict. *H2 is not supported.* Mid-level leagues therefore do not function as disproportionate structural bridges.

7.3 H3 – Community structure

The leiden community detection reveals 17-communities on the aggregated graph with a restriction of at least five leagues. The communities are geographically coherent. The labels designed by hand include *Italian* (■), *British Isles* (■), *German–Austrian* (■), *Francophone* (■), *Iberian* (■), *Pan-American* (■), *Lusophone & East Asian* (■), *Post-Soviet (CIS)* (■), *Balkan & Eastern Mediterranean* (■), *Turkish* (■), *Nordic–Dutch* (■), *MENA & African* (■),

Swiss (■), East Asian & Pacific (■), and Czecho-Slovak (■). See Figure 3 for a global and 4 for a European map.

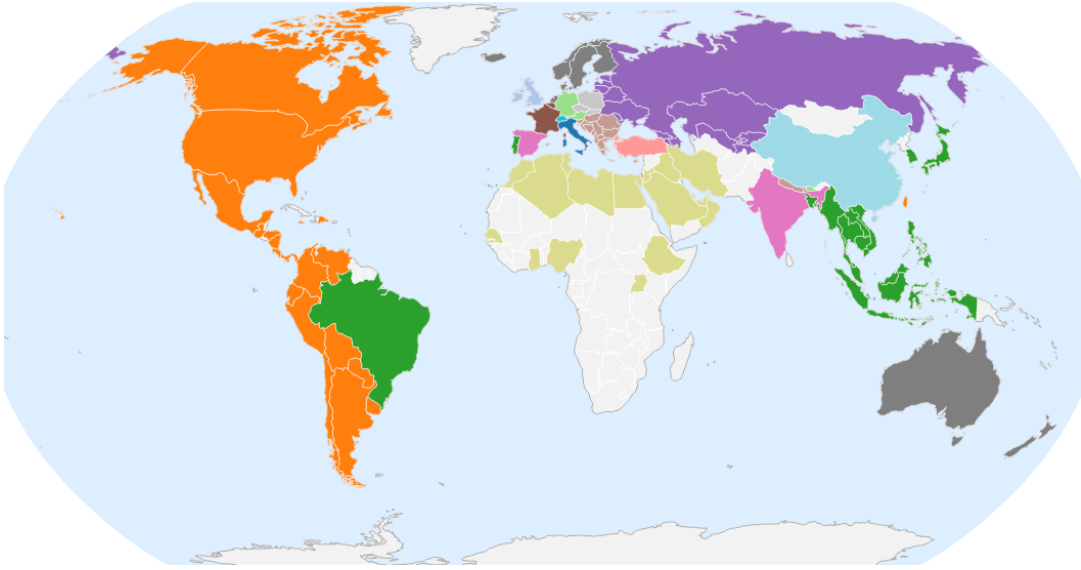


Figure 3: Global Football Transfer communities

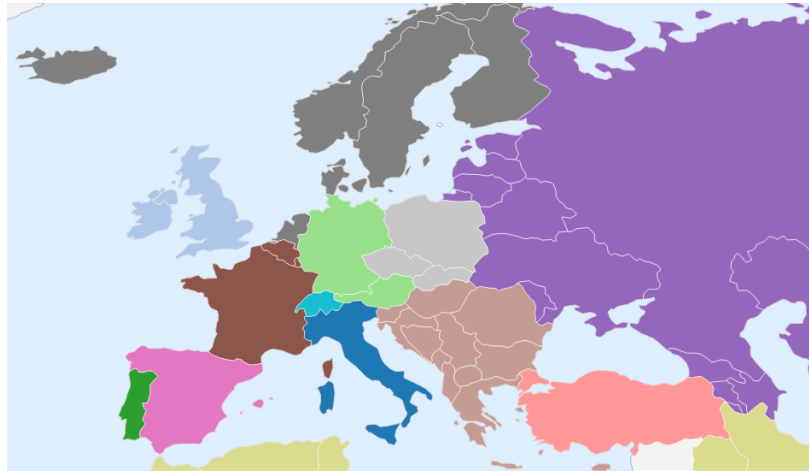


Figure 4: European Football Transfer communities

The null-model significance test compares the observed modularity 20 degree-preserving random rewirings of the self-loop-free graph (Section 6):

- Observed modularity $Q^{\text{obs}} = 0.2726$
- Null mean modularity $\bar{Q}^{\text{null}} = 0.0681$
- Standardised deviation $z = 169.55$
- Empirical one-sided $p = 0.000$ (no null run exceeded Q^{obs})

The per-season analysis shows that in every one of the 26 seasons examined from 2000/01 onwards, observed Q exceeds the null mean.

Verdict. H3 is *supported*. The transfer network has strong, non-random community structure that is stable across seasons and geographically interpretable.

7.4 H4 – Lower-league pathways

The transfer-count-weighted domestic share of outgoing transfers increases with source-league tier:

- Top (Tier 1): 56.3% domestic
- Second (Tier 2): 77.8% domestic
- Mid (Tier 3–4): 91.7% domestic
- Lower (Tier 5+): 91.9% domestic

Both mid- and lower-tier shares exceed the 70% threshold by a wide margin (Figure 5). 56.5% of transfers originating in Tier 5+ source leagues are upward (to a higher tier).

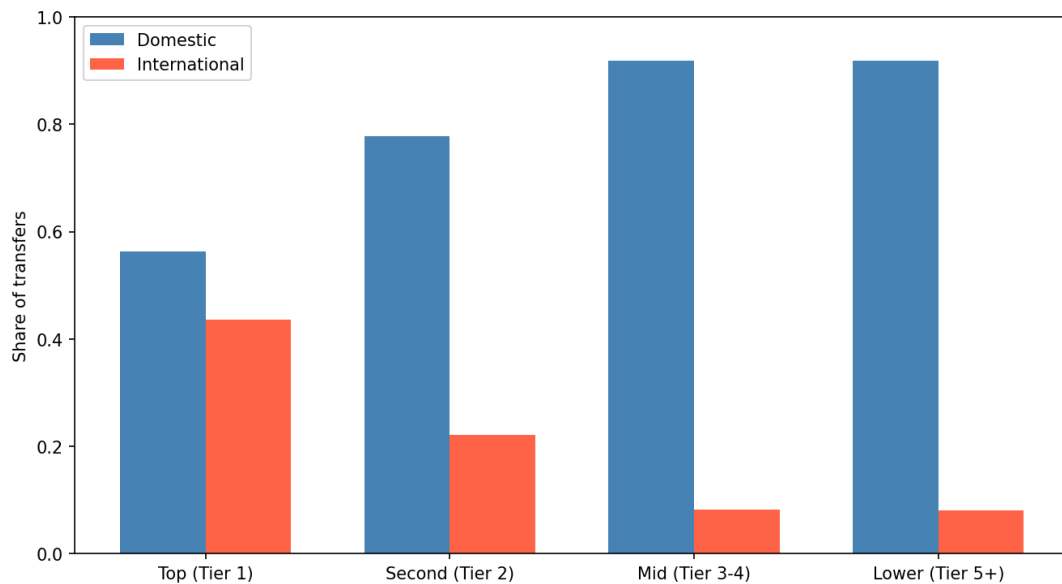


Figure 5: Domestic vs International Transfer Share by Source Tier

Verdict. H4 is *supported*. Lower-tier leagues are a national labor market compared to higher-tier leagues.

7.5 Summary against KPIs

H1 – Concentration.

NOT SUPPORTED (DECLINING TREND).

Threshold: $HHI > 0.15$ or $Top-5 > 50\%$.

Result: $HHI = 0.0094$, $Top-5 = 12.3\%$. Mann–Kendall: strong monotonic decline ($\tau = -0.93$, $p < 10^{-4}$).

H2 – Bridge leagues.

NOT SUPPORTED.

Threshold: $lift \geq 1.5\times$.

Result: Mid-tier share in top-25 betweenness = 40.0% vs. baseline 40.3%; $lift = 0.99\times$.

H3 – Community structure.

SUPPORTED.

Threshold: $p < 0.05$.

Result: $Q^{\text{obs}} = 0.273$ vs. $\bar{Q}^{\text{null}} = 0.068$; $z = 169.6$, $p = 0.000$.

H4 – Lower-league pathways.

SUPPORTED.

Threshold: domestic share $> 70\%$ for tier 3+.

Result: Tier 3–4 = 91.7% domestic, Tier 5+ = 91.9% domestic.

8 Deployment

The final result of this project is a reproducible pipeline. This section therefore presents the produced results, where to find them in the project and how to reproduce them end-to-end. The complete project is available at <https://github.com/timg4/Mobility-in-Global-Football>.

The pipeline itself is split into two notebooks `01_data_preparation.ipynb` and `02_analysis.ipynb` that are to be executed in sequence. When done so, they produce the cleaned edge-list CSVs, the result-figures and two final summary tables.

The Reproducibility is supported by having pinned Python dependencies; a fixed random seed with `seed = 12412672` for all random experiments such as the Leiden partitioning; and both notebooks use relative paths to the project root, so rerunning them would produce the exact same figures and numbers.

Three figures are additionally produced as interactive HTML files. The community-metagraph can be explored by hovering over the communities which reveals its league members and the size of each cross-community flow. The community-map is also represented in an HTML as a choropleth of the dominant countries per community. The static PNGs embedded in the report show the overall shape, while these interactive visualisations allow a drill down into individual leagues and corridors.

9 Conclusion and Outlook

This project created a directed, weighted league-to-league network over 32-seasons, built from 586,520 league-to-league transfer records across 672 leagues in 118 countries and used it to test four hypotheses.

Two of these hypotheses could be supported by the data and two were not supported. The network is not concentrated in the sense of they proposal’s thresholds ($\text{HHI} = 0.0094$, $\text{Top-5 share} = 12.3\%$) and mid tier-leagues actually do not function as structural bridges between leagues ($\text{lift} = 0.99\times$ vs. baseline). By contrast, the market shows strong, geographically coherent community structure ($Q = 0.273$, $z = 169.6$, $p = 0.000$ against a degree-preserving null model) that is stable across season. The found communities map clearly into linguistic, regional and especially historical ties. The data also suggests that Lower-tier leagues behave overwhelmingly as national labour markets, with domestic transfer shares above 91% for tier 3 and below. Another also very interesting finding is that inflow concentration has declined significantly over the entire observation window ($\tau = -0.93$, $p < 10^{-4}$), which amplifies the globalization of the player labour market.

Limitations

This work has several limitations. Transfers fees were excluded from the analysis because of their non-random missingness. A analysis restricted to leagues with reliable fee coverage would support the count-based view used here, particularly for the economically strong top-tier European leagues. In addition to that, loans were counted as ordinary transfers although they are exceptionally different from permanent. While a distinct analysis was done, it is not included in these results. Also the H3 null distribution was done only from 20 degree-preserving rewirings instead of the originally planned 1,000 due to computational limitations. A larger rerun would tightens the empirical p -value.

Outlook

Future work could combine the league-level network from this project and combine it with club-level networks of prior work (???). This would result in a two-layer model in which clubs are nested withing their league communities and league-level flows constrain club-level mobility. Another direction would be to incorporate player attributes such as nationality, age or position to get structural findings on which specific players are moving. Another very natural follow-up would be to include fees in the analysis, as mentioned above.